

Deep Learning in Linguistics and in Computer Science



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A Comment on "The Implications of Neural Networks and Other Learning Models for Linguistic Theory," Special Issue of "Linguistics Vanguard" 12(1), Guest Editors: Vsevolod Kapatsinski and Gasper Begus,

<https://www.degruyterbrill.com/journal/key/lingvan/12/s1/html>

As a reviewer of three papers submitted to this special issue, I feel obliged to react. I already commented on my experience with the issue last year in *Modeling Language Without Language: A ChatGPT Lesson for Language Research*

(<https://ling.auf.net/lingbuzz/0089998>), warning the community that the papers assume words where there are none. LLMs work with tokens, and owing to an unfortunate coincidence of terminology, many linguists seem to believe that LLM tokens are words. Yet LLM tokens are not linguistic units. Thank you for the 4,889 downloads of the paper as of July 25, 2026. It is good to know that so many linguists took the trouble to read my text.

Having now looked at the published collection of papers and, in particular, at the references in the introductory article, I am surprised that this special issue was published in its present form. The literature to which the editors refer largely predates recent developments in language learning in LLMs—developments that have fundamentally changed NLP and consequently carry significant implications for linguistic theory.

I understand that, now that the task of fluent language generation has already been solved by computer scientists, compiling a volume on methods for language learning (actually, on language generation) is a real challenge, because:

1. The learning methods are largely computational and not linguistic; the most modern methods, such as those found in LLMs, do not use words or semantics, which is a huge problem for linguistic theory.
2. A conclusion such as “LLMs render linguistic theory irrelevant,” though correct, would be unacceptable to linguists.
3. It is so tempting to claim that linguists have influenced the developments in NLP that led to the rise of LLMs.

Nevertheless, the impression that this special issue leaves is exactly the one listed under point 2: “LLMs render linguistic theory irrelevant.”

Leaving aside neural networks for the moment, let me start with the highly ambitious Naive Discriminative Learning (NDL) framework, which was developed by a linguist; that is, it is not a computer science model and, as a genuinely linguistic product, can save the field’s reputation by demonstrating a similarity in its development to LLMs. Yet NDL is full of assumptions that directly contradict how LLMs operate. This begins with the basic fact that NDL operates on sequences of two or three letters (bigrams and trigrams) recognized within words and associated with lexical cues, whereas LLMs view language as an uninterrupted sequence of tokens divorced from semantics.

Furthermore, NDL, as demonstrated in the special issue, treats “deep” and “linear” as incompatible features, whereas LLM architectures incorporate both simultaneously. But, obviously, papers by Harald Baayen and colleagues cannot simply be set aside.

Intriguingly, although LLMs use linear sequences of data, do not differentiate between derivation and inflection, and include no paradigms, syntactic trees, or relational

networks of any type, linguists find implications for maintaining such structures in linguistic analyses.

As for child language learning and comparisons with LLMs, I can only hope that, at some point, we will be offered papers that interpret language learning without manipulative preliminary assumptions.

Another sad fact revealed by the special issue is that the machine-learning models (neural networks) discussed in this collection, although originating in computer science, have been treated as linguistic theories; that is, linguists have supplied words as input and evaluated the correctness of the resulting outputs. The special issue therefore illustrates a broader problem: despite decades of linguistic research on neural networks and other learning models, linguistic work on the topic remains largely out of step with contemporary NLP systems. The methods that transformed NLP emerged from an entirely different research tradition.

For a comparison of the claims made in this special issue with those of contemporary deep-learning research, see *MIT Introduction to Deep Learning* (6.S191), a free course with editions from different years, available at:

https://www.youtube.com/playlist?list=PLtBw6njQRU-rwp5__7C0oIVt26ZqjG9NI

Special attention should be paid to the lectures on reinforcement learning and LLMs because there is much misinformation about these topics in current linguistic theory and in this special issue.

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